

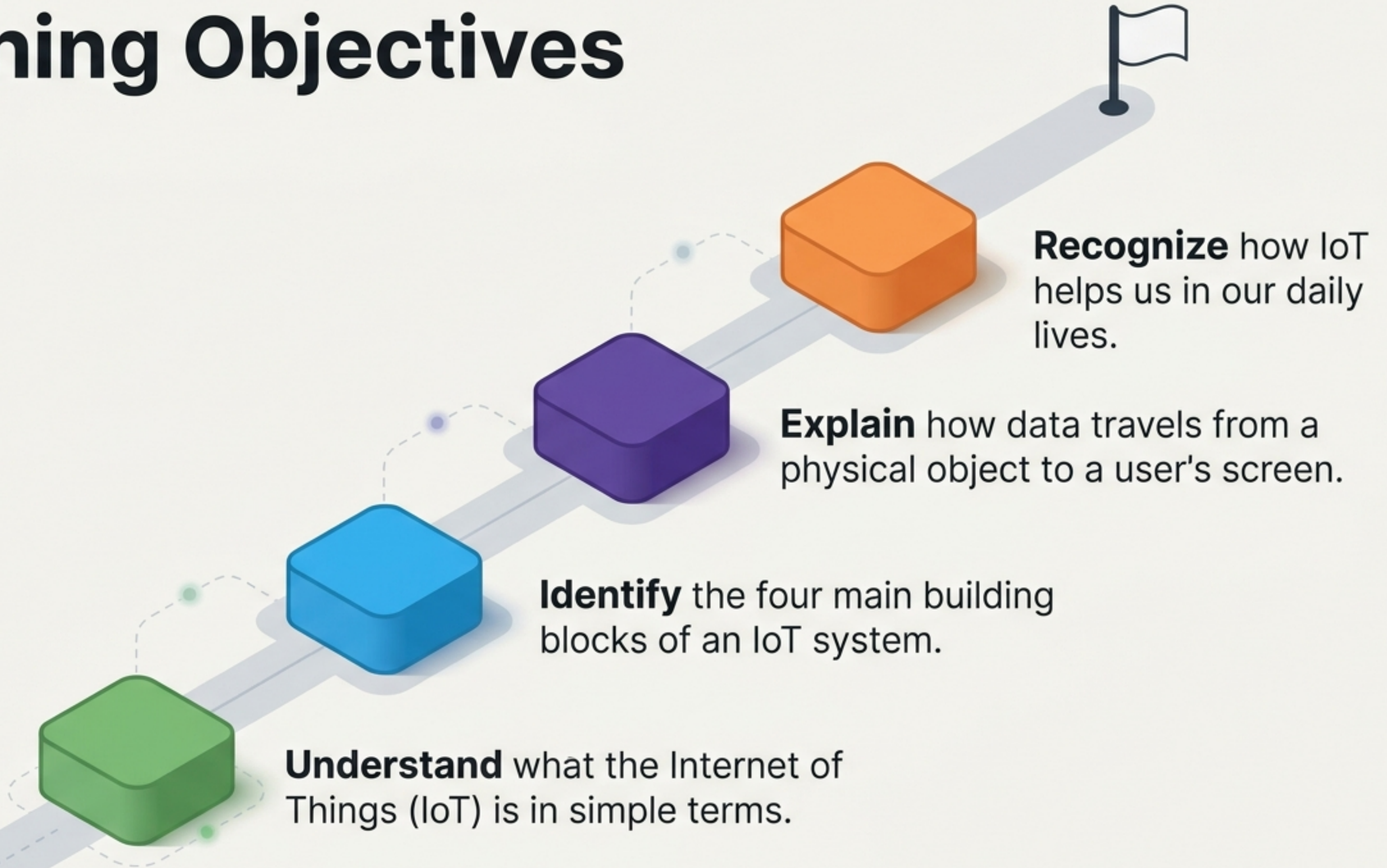
Module 1: Introduction to IoT Systems

IIT67-372 Internet of Things

Lecture Asst. Prof. Dr. Chanankorn Jandaeng
School of Informatics, Walailak University, Thailand



Learning Objectives



What is IoT?

The Big Idea: IoT connects normal, everyday objects to the internet.

- “Things” in IoT are physical objects with small computers inside them.
- These objects can “talk” to each other and share information with us.

The Old Way



Checks the time.

The IoT Way



Checks your heartbeat and sends the data to your phone.

The Senses (Devices)

How IoT 'feels' the world: The first step is gathering information from the physical environment.

Sensors (The Eyes and Ears)

- They collect data from the real world.
- Example: A thermometer measuring if a room is hot or cold.

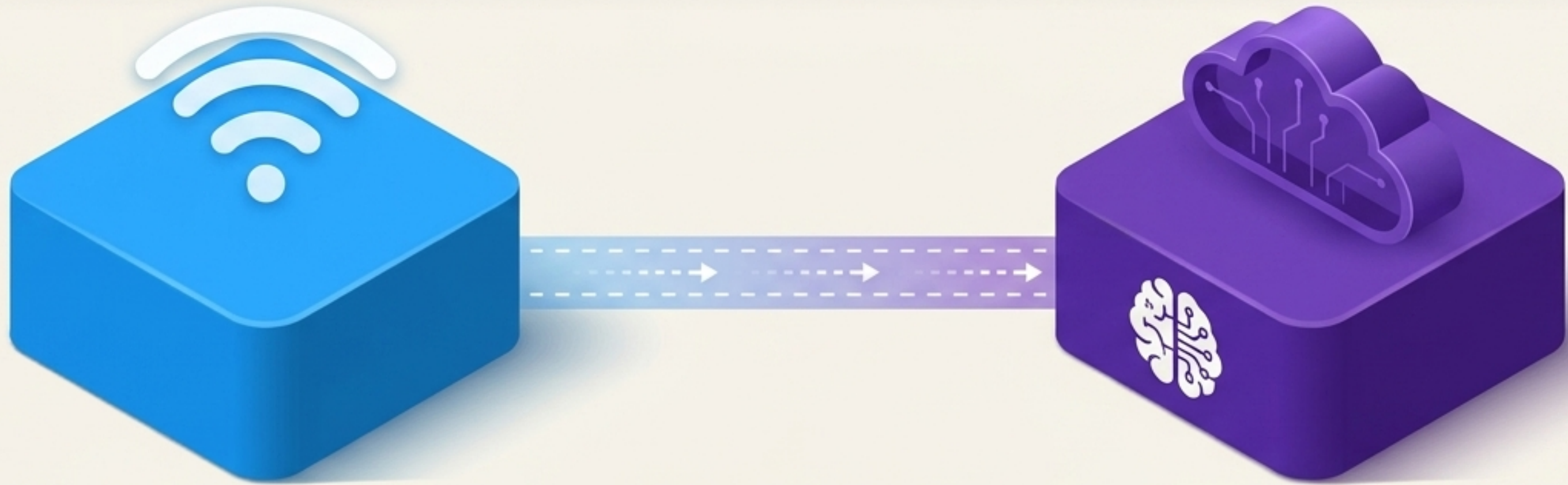
Actuators (The Hands)

- They take action based on instructions.
- Example: A motor turning on a water sprinkler when the ground is dry.



The Path and the Brain (Network & Cloud)

Real-life example: Your home Wi-Fi (Network) sends video from your doorbell camera to an online server (Cloud).



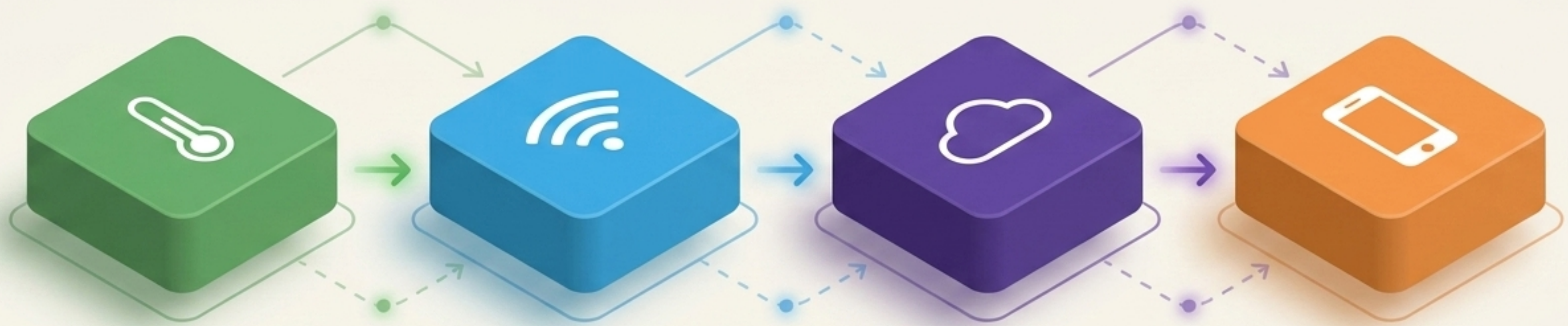
Network (The Path)

- The invisible road that moves data.
- Uses signals like Wi-Fi, cell phone data (4G/5G), or Bluetooth.

Cloud (The Brain)

- A giant, powerful computer system on the internet.
- It safely stores data and uses smart software to understand it.

The Journey of Data



Step 1: Collect

A sensor feels that the soil in a farm is too dry.

Step 2: Send

The sensor uses Wi-Fi to send a message through the air.

Step 3: Think

The internet “brain” receives the message and decides the plant needs water.

Step 4: Show

The farmer gets an alert on their phone screen.

Real-World Application: Smart Farming

What it is: Using IoT to help farmers grow crops better and easier.

How it works: Sensors are placed in the dirt to check soil moisture, temperature, and plant health.

Daily Life Benefit:

- Farmers save water by only watering when needed.
- They grow more food with less work.



Real-World Application: Smart Cities & Environment

Daily Life Benefit: Less time stuck in traffic and cleaner air for everyone to breathe.

Smart Cities: Making towns run smoothly using connected devices.

Example: Traffic lights that know when cars are waiting and change colors to stop traffic jams.



Environment Checkers: Keeping people safe from pollution.

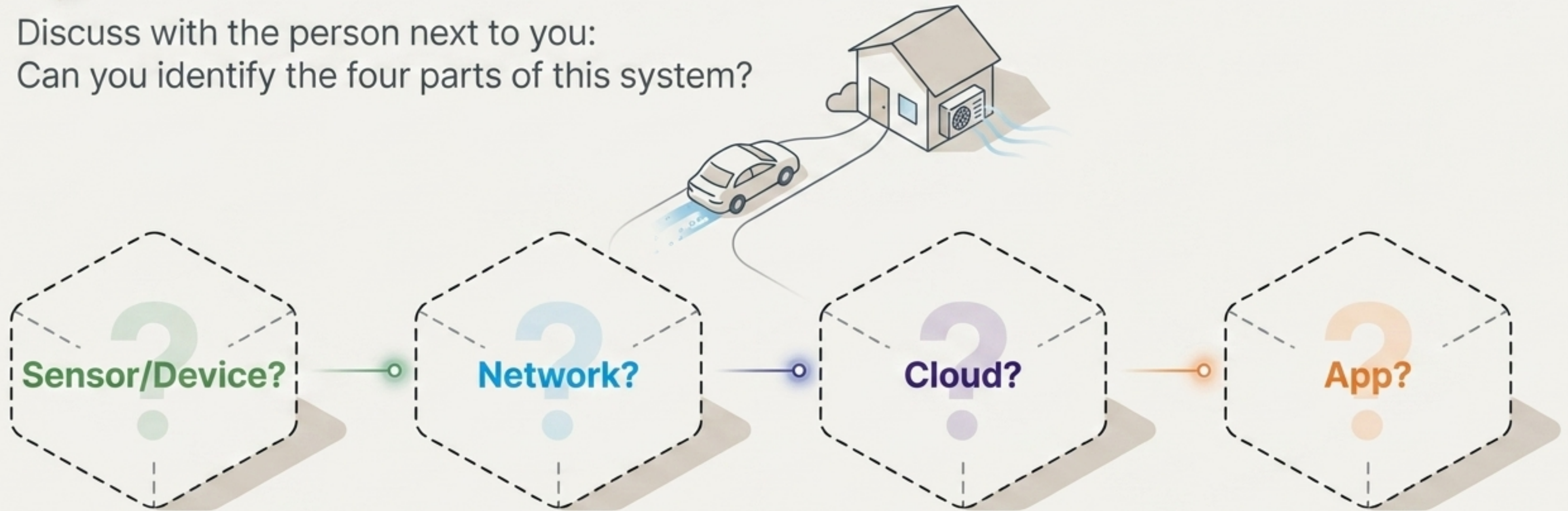
Example: Sensors in the city that measure air quality or check for dirty water.



Activity: Map the Flow!

The Scenario: You are driving home. Your Smart Home Air Conditioner turns on automatically 10 minutes before you arrive so the house is cool.

Discuss with the person next to you:
Can you identify the four parts of this system?



Summary

- IoT connects real-world objects to the digital world to make them “smart.”
- Every IoT system has four main parts: Devices (Sensors/Actuators), Network, Cloud, and Application.
- Data flows in a clear path: it is collected, sent, processed, and then shown to a user.
- IoT is used everywhere—from farming to city streets—to improve our daily lives.



Review Questions

1. What is the simple difference between a sensor and an actuator?

2. Why does an IoT system need the Cloud to work well?

3. Give one example of how a Smart City uses IoT to help the people who live there.

